Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

ULTRAVIOLET SPECTRA OF LiH AND LiD[1]

R. VELASCO[2]

ABSTRACT

The absorption spectra of LiH and LiD have been observed in the near ultraviolet with high dispersion and absorbing path lengths up to 16 meters. A new band system has been found in each molecule involving the ground state and a $^1\Pi$ excited state. Rotational and vibrational analyses of this system have been carried out and rotational and vibrational constants for the upper state have been determined. The observed breaking off of the rotational structure of the bands of this $B\ ^1\Pi$—$X\ ^1\Sigma^+$ system has been interpreted as due to predissociation by rotation. With this assumption very accurate dissociation limits of the $B\ ^1\Pi$ state have been obtained. From these dissociation limits the dissociation energies of the three known electronic states of LiH and LiD have been calculated. In particular the dissociation energies (D_0) of the ground states of LiH and LiD have been found to be 2.4288 ± 0.0002 ev. and 2.4509 ± 0.0010 ev., respectively.

INTRODUCTION

The $A\ ^1\Sigma^+$—$X\ ^1\Sigma^+$ band systems of LiH and LiD have been known for many years, and detailed analyses have been given by Nakamura (1930) and Crawford and Jorgensen (1935). The latter authors were able to use their excellent experimental data to test fine details of the theory of molecular spectra (Crawford and Jorgensen 1936) and to estimate the dissociation energies of the ground states of LiH and LiD. The error in the D_0 values was quite large (±0.2 ev.) because only a few vibrational levels had been observed and the extrapolation from these observed levels to the dissociation limit was rather uncertain.

Since the LiH molecule is a relatively simple one (four electrons) and has been the subject of many theoretical calculations, it would be interesting to have a detailed experimental knowledge to compare with the theory. However, up till now, the $A\ ^1\Sigma^+$ state, which lies 3.3 ev. above the ground state, was the only known excited state of LiH. Since the ionization energy of LiH is presumed to be about 8 ev., many more excited states of LiH are to be expected. If a number of these excited states could be observed experimentally they would yield much interesting information about the electronic structure of this molecule. The present experimental work is an attempt to study some of these higher electronic states of LiH and LiD.

EXPERIMENTAL

The observations were made in absorption. To attain long absorbing paths, an absorption cell equipped with a multiple reflection system of mirrors, of the type first described by White (1942) and modified by Bernstein and Herzberg (1948), was used. Absorption path lengths up to 16 meters were attained.

[1]Manuscript received June 24, 1957.
Contribution from the Division of Pure Physics, National Research Council, Ottawa, Canada.
Issued as N.R.C. No. 4482.
[2]National Research Laboratories Postdoctorate Fellow; present address: Instituto de Optica, Madrid, Spain.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

The absorption cell consisted of a steel tube, 180 cm. long and 5 cm. in diameter, whose ends were attached to short pyrex cylinders in which concave mirrors of 2 meter radius of curvature were mounted. A central section of the tube, 1 meter long, was thermally insulated. This central part was electrically heated by connecting it directly across the secondary of a large step-down transformer whose primary voltage could be controlled by means of a variac. In this way, operating temperatures between 800° and 1000° C. could be easily obtained by altering the voltage in the primary, and could be maintained to within ±25° C. The rest of the tube was cooled by means of water jackets at both sides of the hot section.

Pure lithium metal was placed in the center of the tube and the system was evacuated for several hours with the central part heated to about 300° C. The tube was then heated to a higher temperature and H_2 or D_2 was admitted into the tube until the reaction between the gas and the metal stopped. An excess of several cm. Hg pressure of hydrogen was maintained in the tube to prevent a too rapid diffusion of the hydride to the cooler parts of the system. A high pressure Xe-lamp was used as the source for the continuous radiation.

The region between 2000 and 3200 Å was investigated. The final plates were obtained in the third order of a 21 ft. grating spectrograph giving a dispersion of 0.85 Å/mm. The spectra were photographed on 103a-0 Eastman Kodak plates. In order to eliminate overlapping orders in the regions where the bands appear, a Corning-glass filter No. 7-54 was used, sometimes in combination with a nickel sulphate filter (Kasha 1948). Exposure times of about 2 hours were required with the mirrors set to give a path length of 16 meters.

The measurements were made against a Fe-arc comparison spectrum, using as standards the Fe wave lengths given by Harrison (1939) and Edlén (1955). Each band was measured on two different plates with a Gaertner comparator, and the wave lengths in air obtained in this way were reduced to wave numbers by means of Edlén's vacuum correction tables (Edlén 1952).

RESULTS AND ANALYSIS

A new band system, observed both in LiH and in LiD, was identified in the region 2880–3080 Å. The new system consists of two groups of bands degraded to the red and separated by about 1500 cm^{-1} in LiH and 1000 cm^{-1} in LiD. The individual bands in each group do not stand out very clearly because the rotational structure is very open and the vibrational intervals in the upper state turn out to be rather small. All bands show a clear breaking off of the rotational structure. As will be discussed more fully below, this breaking off can be attributed to rotational predissociation of the upper state.

Plates taken with a low dispersion instrument also show a very strong continuum on the short wave length side of these new bands with maxima of absorption at about 2720 Å (LiH) and 2690 Å (LiD). From the results of our analysis, both the position and the intensity distribution of this continuum agree well with what could be expected if the continuum corresponds to transitions from the ground state to the steep part of the potential curve of the new excited state above the dissociation limit.

Contrary to our expectations no Rydberg series of band systems have been found in the region investigated. The observation of discrete bands below 2800 Å becomes difficult owing to the presence of the strong molecular continuum just mentioned and the still stronger ionization continuum of atomic lithium which begins at 2300 Å. Experiments in the region 1500–2000 Å might be more successful in this respect.

In Tables I and II the wave numbers of the observed lines belonging to the new systems have been listed. Since the mean difference between the wave numbers of the same line obtained from two different plates is ± 0.04 cm^{-1} we estimate that the wave numbers presented in these tables are probably correct within these limits.

The observed differences between the two groups of bands in each system roughly correspond to the known vibrational frequencies of the ground states of LiH and LiD and the bands in each group may therefore be assigned to v' progressions. Each band consists of an intense Q branch, a less intense R branch, and a still weaker P branch, thus indicating that the band system arises from a transition in which $\Delta\Lambda = \pm 1$. The fact that the lower state of the transition is, as expected, the well-known ground $^1\Sigma$ state of the molecule is confirmed by the agreement of the combination differences $\Delta_2 F''(J)$ with the

TABLE I

WAVE NUMBERS OF THE LINES IN THE $B\,^1\Pi - X\,^1\Sigma^+$ SYSTEM OF LiH

	0–0 band			1–0 band			2–0 band		
J	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$
0			34318.03 (4)			34446.58 (4)			34490.12 (1h)
1		34303.21 (6)	314.68 (5)		34431.76 (5)	438.86 (5)		34475.33 (1h)	477.87 (1)
2	34273.67 (2h)[a]	285.05 (8)	302.06 (5)	34402.21 (2)	409.24 (6)	419.42 (4)	—	448.25 (2h)	
3	240.71 (3)	257.62 (8)	279.95 (5)	364.88 (3)	375.07 (8)	387.91 (4)	34403.89 (0)		
4	198.66 (5)	220.86 (8)	248.06 (5)	316.05 (3)	328.89 (5)	343.48 (2)			
5	147.26 (5)	174.36 (8)	205.98 (6)	255.28 (2)	269.77 (4)				
6	086.24 (4)	117.73 (8)	152.98 (3)	181.63 (2)					
7	34015.32 (5)	34050.34 (6)	088.01 (2)						
8	33933.65 (3h)	33971.10 (4)							
9	840.46 (1h)								

	0–1 band			1–1 band			2–1 band		
J	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$
0			32958.31 (2)			33086.96 (2h)			33130.7[c](0)
1		32943.75[b](10)	955.37 (3)		33072.50 (4)	079.62 (3)		33116.0 [c](0)	118.7[c](0)
2	32915.26 (1)	926.64 (5)	943.75[b](10)	33043.93 (1)	050.85 (5)	061.03 (3)	—	089.93 (1)	
3	883.60 (3)	900.53 (5)	922.84 (4)	33007.73 (2)	33017.98 (5)	33030.81 (4)	—		
4	843.27 (4)	865.47 (5)	892.61 (4)	32960.63 (2)	32973.39 (5)	32988.13 (3)			
5	793.84 (2)	821.06 (6)	852.66 (4)	901.95 (2)	916.43 (2)				
6	735.38 (1)	766.86 (5)	802.08 (3)	830.87 (1h)					
7	667.25 (2)	702.50 (2)	740.18 (1h)						
8	589.02 (1)	626.60 (3)							
9	499.29 (1)								

[a] Eye-estimate relative intensities. h: diffuse?
[b] Line common to two bands or branches.
[c] It is doubtful if these lines belong to the system. They have not been used in the calculation of constants.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

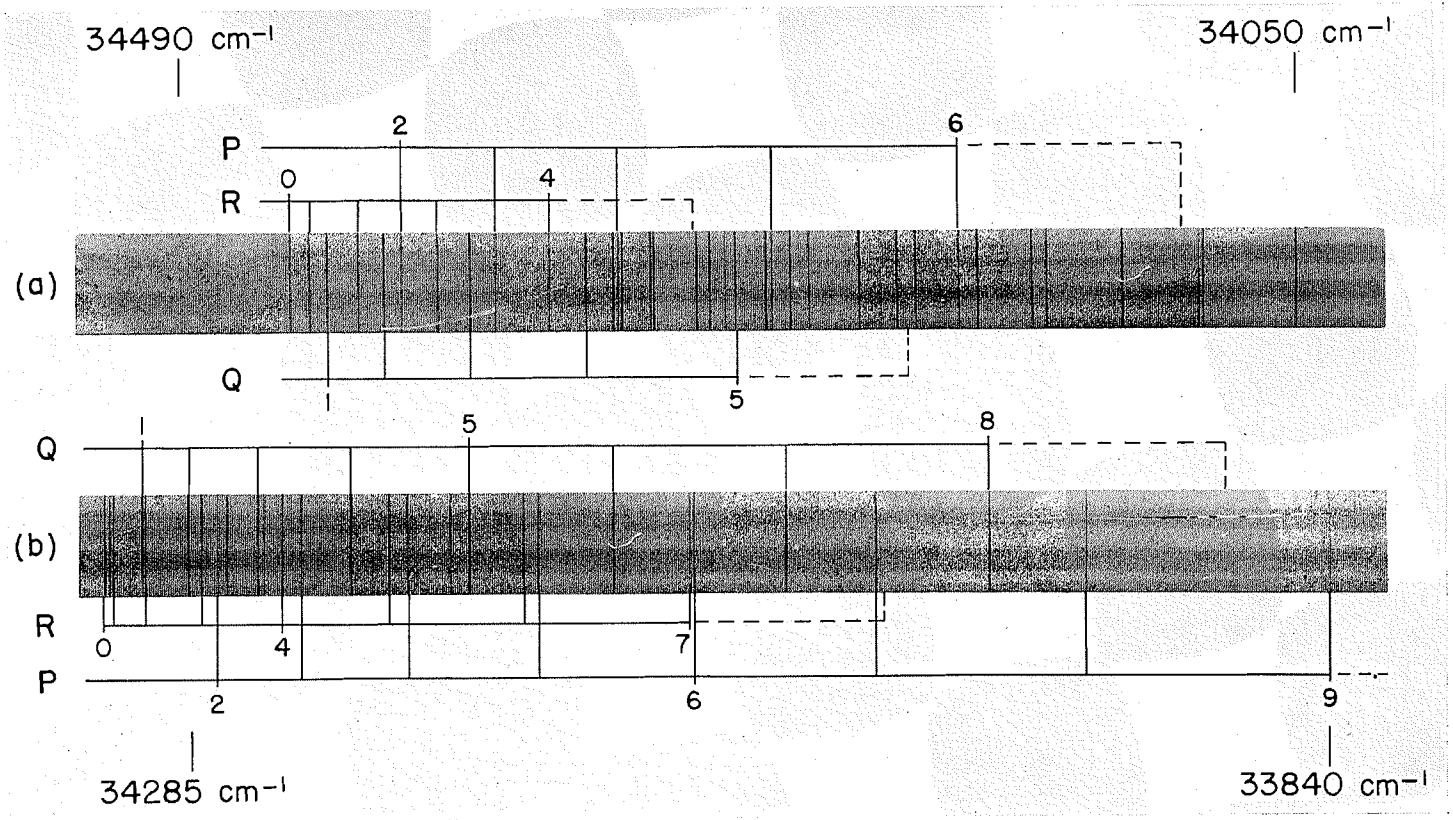


FIG. 1. The $v'' = 0$ progression of the $B\,^1\Pi$—$X\,^1\Sigma^+$ system of LiH. (*a*) Rotational structure of the 1–0 band. (*b*) Rotational structure of the 0–0 band. A few very weak lines on the left-hand side of the 1–0 band belong to the 2–0 band. The broken lines indicate the position of the first missing line in the corresponding branch.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

TABLE II

WAVE NUMBERS OF THE LINES IN THE $B\ ^1\Pi$—$X\ ^1\Sigma^+$ SYSTEM OF LiD

	0–0 band			1–0 band			2–0 band		
J	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$
0			34469.59 (1)			34587.86 (1)			34647.59 (0)
1		34461.27[b](4)	467.98 (2)		34579.39 (2)	584.31 (3h)		34639.30 (1h)	642.46 (0h)
2	—	451.24 (2h)	461.27[b](4)	34562.56 (0)	567.58 (3)	575.00 (2)	—	625.68 (2h)	630.36 (0h)
3	34426.22 (3)	436.23 (3)	449.57 (2)	542.42[b](4)	549.87[b](5)	559.55 (4)	—	605.12 (3h)	611.01 (1h)
4	402.91 (1)	416.11[d](5)	432.70 (3)	516.50 (2h)	526.02 (5)	537.83 (3)	34571.95 (0)	577.37 (2)	584.31 (3h)
5	374.47 (2h)	390.88 (5)	410.46 (1)	484.33 (2h)	496.07 (5)	509.68 (2h)	535.74 (0)	542.42[b](4)	549.87[b](5)
6	340.95 (3)	360.35 (3)	382.86 (2)	446.04 (2)	459.61 (4)	474.97 (2h)	492.71 (1)	499.52 (2h)	
7	302.10 (2h)	324.45 (4)	349.70 (1)	401.51 (1)	416.54[d](5)	432.70 (3)	—		
8	257.94 (3h)	283.05 (3)	310.63 (2h)	350.19 (1)	366.39 (3)	382.86 (2)			
9	208.56 (2)	235.89 (3)	265.41 (0h)	291.93 (1)	308.21 (3)				
10	153.20 (2)	182.58 (3)	213.77 (0)	225.62 (0)					
11	091.90 (0h)	122.84 (2h)	154.41 (1)						
12	024.50 (1)	055.71 (0)							
13	—								

	0–1 band			1–1 band			2–1 band		
J	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$	$P(J)$	$Q(J)$	$R(J)$
0			33440.50 (2h)			33558.79 (2)			33618.65 (1)
1		33432.51 (2)	439.21 (2)		33550.60 (5)	555.54 (3)		33610.43 (2)	613.59 (2)
2	33416.06 (0)	422.78 (3)	432.87 (2)	33534.26 (1h)	539.20 (6)	546.53 (3)	33594.19 (0h)	597.19 (4)	601.69 (3)
3	398.23 (1)	408.24 (4)	421.61 (3)	514.71 (2h)	522.01 (3)	531.59 (3)	572.52 (1h)	577.14 (4)	582.97 (3)
4	375.62 (2)	388.89 (5)	405.41 (4)	489.27 (3)	498.88 (5)	510.59 (4)	544.46 (1)	550.32 (5)	556.97 (2)
5	348.08 (3)	364.53 (5)	384.07 (4)	458.10 (4)	469.77 (5)	483.41 (4)	509.46 (1)	516.12 (3)	523.31 (1)
6	315.63 (3)	335.09 (5)	357.51 (3)	420.85 (3)	434.43 (6)	449.60 (4)	467.24 (1)	474.19 (1)	
7	278.16 (3)	300.48 (5)	325.54 (3)	377.39 (4)	392.61 (5)	408.86 (1)	417.38 (1)		
8	235.50 (3)	260.48 (5)	287.97 (2)	327.62 (3)	343.81 (4)	360.31 (1h)			
9	187.55 (3)	214.96 (5)	244.45 (2)	270.88 (1)	287.29 (2)				
10	134.12 (4)	163.44 (3)	194.61 (1)	206.48 (1)					
11	074.74 (2)	105.62 (3)	137.35 (0)						
12	33009.19 (1)	040.87 (3)							
13	32936.82 (1)								

[a,b,c] All symbols as in Table I.
[d]Line blended with another one and difficult to measure.

values given by Crawford and Jorgensen (1935) for the $v'' = 0$ and $v'' = 1$ levels of the ground state. Since the lower state is a $^1\Sigma$ state it follows that the upper state must be a $^1\Pi$ state. A reproduction of the $v'' = 0$ progression of this $B\ ^1\Pi$—$X\ ^1\Sigma^+$ system of LiH is given in Fig. 1.

The consistency of our measurements and a comparison of our results with those of Crawford and Jorgensen can be seen in Tables III and IV, which contain sets of $\Delta_2F(J)$ values for the ground and excited states of LiH and LiD.

Because of the agreement of our combination differences $\Delta_2F''(J)$ with those of Crawford and Jorgensen, the accurate rotational constants given by these authors for the ground state were adopted and used in our calculations. Rotational constants for the $B\ ^1\Pi$ level were obtained in two ways:

(i) *Bands with eight or more lines in each branch.*—From our observed lines and the $F''(J)$ values obtained by Crawford and Jorgensen for the $v'' = 0$ and $v'' = 1$ levels of the ground state, $F'(J)$ values were calculated, both for the

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

TABLE III

COMBINATION DIFFERENCES $\Delta_2F_0''(J)$ IN THE GROUND STATE OF LiH AND LiD

	LiH		LiD	
J	0–0 band $^1\Pi$—$^1\Sigma$ system	Average $^1\Sigma$—$^1\Sigma$ system*	0–0 band $^1\Pi$—$^1\Sigma$ system	Average $^1\Sigma$—$^1\Sigma$ system*
1	44.36	44.43	—	25.14
2	73.97	73.94	41.76	41.87
3	103.40	103.39	58.36	58.52
4	132.69	132.66	75.10	75.18
5	161.82	161.78	91.75	91.79
6	190.66	190.68	108.36	108.29
7	219.33	219.27	124.92	124.71
8	247.55	247.65	141.14	141.05
9			157.43	157.26
10			173.51	173.39
11			186.27	186.36
Mean diff.	±0.05 cm^{-1}		±0.12 cm^{-1}	

*Data from Crawford and Jorgensen (1935).

TABLE IV

COMBINATION DIFFERENCES $\Delta_2F_0(J)$ IN THE $B\ ^1\Pi$ STATE OF LiH AND LiD

	LiH		LiD	
J	0–0 band	0–1 band	0–0 band	0–1 band
2	28.39	28.49	—	16.61
3	39.24	39.24	23.35	23.38
4	49.40	49.34	29.79	29.79
5	58.72	58.82	35.99	35.99
6	66.74	66.70	41.91	41.88
7	72.69	72.83	47.60	47.38
8			52.69	52.47
9			56.85	56.90
10			60.57	60.49
11			62.51	62.61
Mean diff.	±0.07 cm^{-1}		±0.08 cm^{-1}	

series of rotational levels obtained from the Q branches, and for those obtained from the P and R branches, to obtain two sets of constants. The first differences, $\Delta_1F'(J+\frac{1}{2})$, obey the following relation:

$$(1) \qquad \Delta_1F'(J+\tfrac{1}{2})/2(J+1) = B_v-(2D_v-H_v)(J+1)^2+3H_v(J+1)^4$$

and therefore, by plotting the left-hand side of this formula versus $(J+1)^2$, B_v was found as the intercept on the ordinate axis. Using the value B_v obtained in this manner, $([\Delta_1F'(J+\frac{1}{2})/2(J+1)]-B_v)/(J+1)^2$ was calculated and plotted in turn versus $(J+1)^2$. In each case a straight line could be drawn whose intercept gave $-(2D_v-H_v)$ and whose slope gave $3H_v$.

(ii) *Bands with less than eight lines in each branch.*—A similar method was followed, but this time

$$F'(J)/J(J+1) = B_v-D_vJ(J+1)+H_vJ^2(J+1)^2$$

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

was used as starting formula instead of (1) and $F'(J)/J(J+1)$ was plotted versus $J(J+1)$.

In each case the constants were checked by recalculating the $F'(J)$ values and comparing them with the observed ones.

There is evidence of some Λ splitting in the $B\ ^1\Pi$ state, both in LiH and in LiD, with the Q branches arising from the a levels and the P and R branches from the b levels, but for such short bands the observed combination defects are of the order of magnitude of the experimental errors and no good analysis of this doubling could be made.

The constants obtained as a result of the rotational analysis are given in Tables V and VI.

TABLE V

ROTATIONAL CONSTANTS FOR THE UPPER STATE OF THE $B\ ^1\Pi$—$X\ ^1\Sigma$ BANDS OF LiH and LiD

	B'_v		$D_v\times10^3$		$-H'_v\times10^5$	
Band	B^a	B^b	D^a	D^b	$-H^a$	$-H^b$
			LiH			
0–0	2.880	2.880	2.7	2.7	1.5	1.5
0–1	2.875	2.880	2.4	2.3	1.8	2.0
1–0	1.800	1.807	4.3	5.2	7.0	4.0
1–1	1.800	1.809	4.3	5.2	7.0	4.5
2–0	0.64	0.64	—	—	—	—
			LiD			
0–0	1.691	1.696	0.74	0.78	0.25	0.25
0–1	1.691	1.692	0.76	0.76	0.24	0.24
1–0	1.252	1.255	1.35	1.35	0.54	0.53
1–1	1.254	1.253	1.40	1.40	0.54	0.50
2–0	0.799	0.804	2.01	1.98	1.54	1.58
2–1	0.801	0.807	1.91	2.16	1.99	1.47

[a]From Q branches.
[b]From P and R branches.

TABLE VI

MEAN ROTATIONAL CONSTANTS FOR THE VIBRATIONAL LEVELS OF THE $B\ ^1\Pi$ STATE OF LiH AND LiD

Constant	Molecule	$v=0$	$v=1$	$v=2$
B_v	LiH	2.879	1.804	0.64
	LiD	1.693	1.254	0.803
$D_v\times10^3$	LiH	2.6	4.8	—
	LiD	0.76	1.38	2.02
$-H_v\times10^5$	LiH	1.7	5.6	—
	LiD	0.25	0.53	1.65

The observed B_v, D_v, and H_v values of the $B\ ^1\Pi$ state can be represented by the following formulae:

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

$$\text{LiH}\quad\begin{cases} B_v = 3.383-0.986(v+\tfrac{1}{2})-0.045(v+\tfrac{1}{2})^2 \\ 10^3\times D_v = 1.5+2.2(v+\tfrac{1}{2}) \\ 10^5\times H_v = 0.3-3.9(v+\tfrac{1}{2}) \end{cases}$$

$$\text{LiD}\quad\begin{cases} B_v = 1.908-0.427(v+\tfrac{1}{2})-0.006(v+\tfrac{1}{2})^2 \\ 10^3\times D_v = 0.44+0.63(v+\tfrac{1}{2}) \\ 10^5\times H_v = 0.13-0.56(v+\tfrac{1}{2})-0.42(v+\tfrac{1}{2})^2 \end{cases}$$

Not enough information has been obtained for the $B\ ^1\Pi$ state of LiH and LiD to attempt a correction of these constants in the same way that was done by Crawford and Jorgensen for the ground state and the $A\ ^1\Sigma^+$ state.

The results of the vibrational analysis are presented in Table VII, which contains Deslandres arrays of the band origins of the observed bands of LiH and LiD. The vibrational numbering for the ground state follows immediately

TABLE VII

DESLANDRES TABLES OF THE $B\ ^1\Pi$—$X\ ^1\Sigma^+$ BANDS OF LiH AND LiD

	LiH			LiD		
v'	$v''=0$		$v''=1$	$v''=0$		$v''=1$
0	34312.26	(1359.64)	32952.62	34466.27	(1029.03)	33437.24
	130.73		130.72	119.00		119.05
1	34442.99	(1359.65)	33083.34	34585.27	(1028.98)	33556.29
	45.9		46.0	60.82		60.73
2	34488.9	(1359.6)	33129.3	34646.09	(1029.07)	33617.02

from the rotational analysis, but for the upper state the numbering is not so obvious. The numbering proposed has been adopted because it is the only one for which the isotopic relations for the rotational and vibrational constants are fulfilled. With this numbering the observed vibrational levels of the $B\ ^1\Pi$ states can be represented by the following expressions:

$$\text{LiH}: G_v = 215.5(v+\tfrac{1}{2})-42.4(v+\tfrac{1}{2})^2$$
$$\text{LiD}: G_v = 177.28(v+\tfrac{1}{2})-29.13(v+\tfrac{1}{2})^2$$

The rotational and vibrational constants of LiH and LiD in the $B\ ^1\Pi$ state, which have been determined in the present investigation, are summarized in Table VIII. For comparison, the constants of LiD calculated from those of LiH by means of the isotopic conversion factors ρ^n (where $\rho = (\mu_{\text{LiH}}/\mu_{\text{LiD}})^{\frac{1}{2}}$) given by the elementary isotope theory (Herzberg 1950) are given in the last column. From the values in this table it can be seen that in spite of the large amount of anharmonicity the isotopic relations hold fairly well, especially for those constants that have been most accurately determined (B_e and α_e).

DISCUSSION

The observed breaking off of the rotational structure of the $B\ ^1\Pi$—$X\ ^1\Sigma^+$ bands of LiH and LiD occurs at values of J' which are the same for all the branches of a particular band and decrease with increasing v', thus proving that the perturbed state must be the upper state. That this perturbation can be

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

TABLE VIII

MOLECULAR CONSTANTS OF THE $B\ ^1\Pi$ STATE OF LiH AND LiD (CM^{-1})

Constant	LiH obs.	LiD obs.	ρ^n*	LiD calc. = LiH$\times\rho^n$
B_e	3.383	1.908	$\rho^2 = 0.563135$	1.905
$D_e\times10^3$	1.5	0.44	$\rho^4 = 0.317121$	0.48
$H_e\times10^5$	0.3	0.13	$\rho^6 = 0.178582$	0.05
α_e	0.986	0.427	$\rho^3 = 0.422589$	0.417
γ_e	0.045	0.006	$\rho^4 = 0.317121$	0.014
$\beta_e\times10^3$	2.2	0.63	$\rho^5 = 0.237975$	0.52
$\delta_e\times10^5$	3.9	0.56	$\rho^7 = 0.134012$	0.52
ω_e	215.5	177.28	$\rho = 0.750423$	161.7
$\omega_e x_e$	42.4	29.13	$\rho^2 = 0.563135$	23.9
T_e	34912	34909	—	—
r_e (Å)	2.378	2.376		

*Atomic masses used: $H^1 = 1.008131$, $H^2 = 2.014725$, $Li^7 = 7.018163$ (Mattauch 1946).

attributed to predissociation by rotation[3] is deduced by plotting the energies of the last observed and the first missing rotational levels of the upper state as a function of $J'(J'+1)$. Assuming that for each vibrational level the real breaking off points lie between the last observed and the first missing rotational level, the limiting curve of dissociation (Fig. 2) was drawn. The shape of this curve, with horizontal tangent at $J'(J'+1) = 0$, corresponds to that which

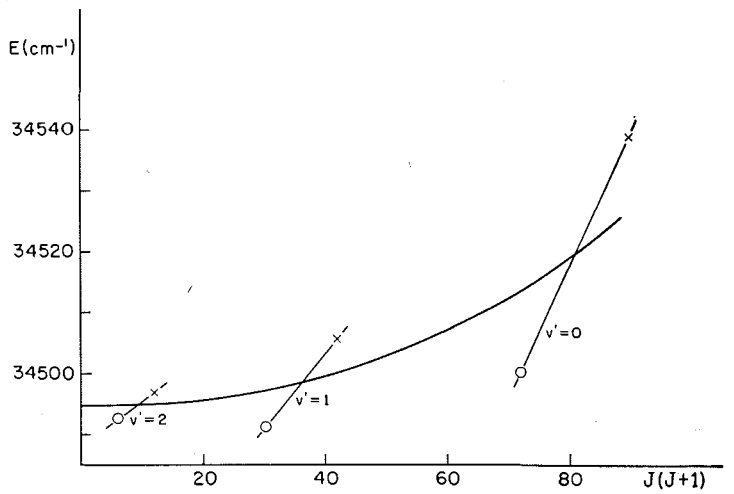


FIG. 2. Limiting curve of dissociation for the $B\ ^1\Pi$ state of LiH.
○ = Last observed rotational levels.
× = First missing rotational levels.

would be expected from a rotational predissociation in which the potential curve of the rotationless state has no maximum (or if any maximum exists it could only be at a very large internuclear distance and therefore extremely small). Furthermore this appears to be the only possible explanation for the

[3]Case III of predissociation (Herzberg 1950).

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

observed predissociation. If it were due to overlapping with the continuum of either of the two lower $^1\Sigma^+$ states the breaking off would be present only in the P and R branches, which is contrary to observation. The remote possibility that the perturbation was due to interaction with the $^3\Sigma$ state, arising from normal Li and H atoms, is excluded by the absence of a maximum in the potential curve of the rotationless state.

A breaking off of the rotational structure of a band system observed in absorption is expected if predissociation occurs in the ground state. What is normally observed when the predissociated state is the upper state is a gradual increase in diffuseness of those lines involving rotational levels above the dissociation limit. The apparent breaking off observed in this particular example could be explained by the fast onset of predissociation (due to the extreme shallowness of the B $^1\Pi$ state), which in some cases does not even allow the observation of levels above the dissociation limit (Fig. 3).

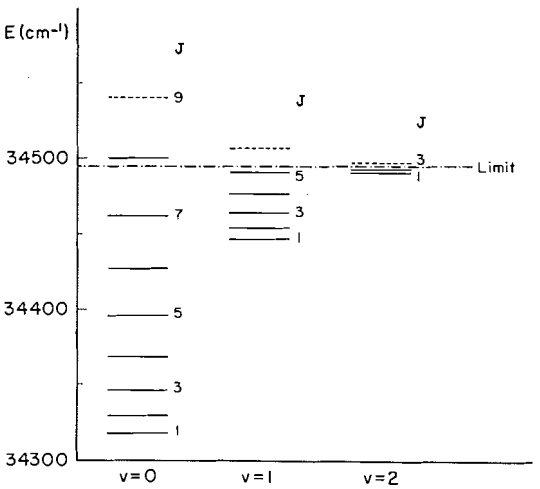


FIG. 3. Rotational levels of the B $^1\Pi$ state of LiH. The broken lines indicate the calculated position of the first missing levels.

The intensity distribution in the rotational structure of the bands B $^1\Pi$—X $^1\Sigma^+$, especially in bands involving levels with $v' = 1$ and $v' = 2$, does not follow the usual pattern deduced from simple considerations about the thermal distribution of rotational levels in the ground state. This effect is not clearly understood but perhaps it might also be related to the fact that the upper electronic state is extremely shallow. It is believed that it does not affect our interpretation of the observed breaking off.

Accepting that the breaking off is due to predissociation by rotation, and that the rotationless potential curves of the B $^1\Pi$ state of LiH and LiD have no maximum, the extrapolation of the limiting curve of dissociation will intersect the ordinate axis at a point giving the dissociation limit. In this particular example the extrapolation is very short and accurate. In the last observed vibrational levels the rotational constant $B_{v'}$ is rather small and

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

in these levels very few stable rotational levels exist below the breaking off point, thus producing very little uncertainty in the extrapolation. The dissociation limits obtained for the $B\ ^1\Pi$ state of LiH and LiD are 34495 ± 2 cm^{-1} and 34673 ± 8 cm^{-1}, respectively.

Because of the simplicity of the electronic structure of the Li-atom it is clear that the only possible dissociation products of this $B\ ^1\Pi$ electronic state of lithium hydride are one hydrogen atom in its ground state and one lithium atom in the $2p$ excited state. With regard to the other two known electronic states we assume that $A\ ^1\Sigma^+$ also dissociates into $H(1s^2S)$ and $Li(2p^2P)$ (Mulliken 1936; Rosenbaum 1938), and that the $X\ ^1\Sigma^+$ state dissociates into normal atoms as is generally admitted. Therefore, from the observed dissociation limits of the $B\ ^1\Pi$ states the following values for the dissociation energies of the three observed electronic states of LiH and LiD are obtained:

$$D_0(B\ ^1\Pi)\begin{cases}\text{LiH: } 183\text{ cm}^{-1} = 0.0227\text{ ev. } (\pm0.0002\text{ ev.})^4\\ \text{LiD: } 207\text{ cm}^{-1} = 0.0257\text{ ev. } (\pm0.0010\text{ ev.})\end{cases}$$

$$D_0(A\ ^1\Sigma^+)\begin{cases}\text{LiH: } 8552\text{ cm}^{-1} = 1.0602\text{ ev. } (\pm0.0002\text{ ev.})\\ \text{LiD: } 8590\text{ cm}^{-1} = 1.0650\text{ ev. } (\pm0.0010\text{ ev.})\end{cases}$$

$$D_0(X\ ^1\Sigma^+)\begin{cases}\text{LiH: } 19591\text{ cm}^{-1} = 2.4288\text{ ev. } (\pm0.0002\text{ ev.})^5\\ \text{LiD: } 19769\text{ cm}^{-1} = 2.4509\text{ ev. } (\pm0.0010\text{ ev.})^5\end{cases}$$

To obtain the corresponding dissociation energies, D_e, referred to the equilibrium position, the zero point energy of the electronic level must be added to each D_0 value. The expression for the zero point energy is

$$E_0 = E_0{}^* + Y_{00} = \tfrac{1}{2}\omega_e - \tfrac{1}{4}\omega_e x_e + \tfrac{1}{8}\omega_e y_e + \ldots + Y_{00},$$

where Y_{00} is the finer correction due to anharmonicity (Dunham 1932). In terms of the observed constants this correction can be expressed as follows:[6]

$$Y_{00} = \frac{B_e}{4} + \frac{\alpha_e\omega_e}{12B_e} + \frac{\alpha_e{}^2\omega_e{}^2}{144B_e{}^3} - \frac{\omega_e x_e}{4}.^7$$

In this way the dissociation energies for the vibrationless states were computed and the results are given in Table IX.

The agreement between the values of D_e for the two isotopic molecules is very good, and supports the interpretation of the breaking off as given above. A systematic shift of 4 or 5 cm^{-1} can be explained by the errors in the determination of the dissociation limits. The remaining differences are mainly due to the uncertainties in the zero point energy calculations. If any real electronic isotope shift of the energy differences $B\ ^1\Pi$—$X\ ^1\Sigma^+$ and $A\ ^1\Sigma^+$—$X\ ^1\Sigma^+$ is present it must be rather small.

[4]The conversion factor 1.239767×10^{-4} has been used to convert cm^{-1} into ev.

[5]In computing these dissociation energies the value $\Delta\nu = 14904$ cm^{-1} for the difference between the $2p$ and $2s$ atomic states of Li has been used (Moore 1949).

[6]This expression is obtained directly from Dunham's formulae. In Herzberg's book (Herzberg 1950) the factor $\frac{1}{4}$ in the last term is inadvertently omitted.

[7]Owing to a numerical error in the calculation of Y_{00}, Crawford and Jorgensen (1936) did not apply this correction to their calculation of the term values of the $A\ ^1\Sigma^+$ state of LiH and LiD. Applying it, the values obtained are: $A\ ^1\Sigma^+$ (LiH) = 26509.6 and $A\ ^1\Sigma^+$ (LiD) = 26510.4.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.

TABLE IX

DISSOCIATION ENERGIES OF THE THREE KNOWN ELECTRONIC STATES OF LiH and LiD

State	Molecule	D_0	E_0*	Y_{00}	D_e†	D_e (ev.)	Limit of error‡ (ev.)
$B\ ^1\Pi$	LiH	183	97.2	3.6	284	0.035	±0.001
	LiD	207	81.4	2.2	291	0.036	±0.002
$A\ ^1\Sigma^+$	LiH	8552	123.9	7.5	8683	1.0765	±0.0002
	LiD	8590	93.7	3.8	8687	1.0770	±0.0010
$X\ ^1\Sigma^+$	LiH	19591	697.1	0.8	20289	2.5154	±0.0002
	LiD	19769	524.3	0.4	20294	2.5160	±0.0010

†The values in columns 3, 4, 5, and 6 are in cm^{-1}.
‡Taking into account the uncertainties due to the zero point energy calculation.

ACKNOWLEDGMENTS

The author is deeply indebted to Dr. G. Herzberg for suggesting the problem and for his kind interest during the course of it. Special thanks are due to Dr. A. E. Douglas for his constant advice and encouragement, and to Dr. B. Kleman and Dr. S. R. Polo for many helpful discussions.

REFERENCES

BERNSTEIN, H. J. and HERZBERG, G. 1948. J. Chem. Phys. **16**, 30.
CRAWFORD, F. H. and JORGENSEN, T., JR. 1935. Phys. Rev. **47**, 358, 932.
——— 1936. Phys. Rev. **49**, 745.
DUNHAM, J. L. 1932. Phys. Rev. **41**, 721.
EDLÉN, B. 1952. Vacuum corrections for wave lengths from 2000 to 13500 Å (Lund University, Sweden).
——— 1955. Wave lengths in the iron arc in air (Lund University, Sweden).
HARRISON, G. R. 1939. M.I.T. wavelength tables (John Wiley & Sons, Inc., New York).
HERZBERG, G. 1950. Spectra of diatomic molecules (D. Van Nostrand Co. Inc., Princeton, N.J.).
KASHA, M. 1948. J. Opt. Soc. Am. **38**, 929.
MATTAUCH, J. 1946. Nuclear physics tables (Interscience Publ. Inc., New York).
MOORE, C. E. 1949. Atomic energy levels, N.B.S. Circular 467 (Washington, D.C.).
MULLIKEN, R. S. 1936. Phys. Rev. **50**, 1028.
NAKAMURA, G. 1930. Z. Physik, **59**, 218.
ROSENBAUM, E. J. 1938. J. Chem. Phys. **6**, 16.
WHITE, J. U. 1942. J. Opt. Soc. Am. **32**, 285.

Can. J. Phys. Downloaded from www.nrcresearchpress.com by Depository Services Program on 10/06/14
For personal use only.